## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 1 of 5 · Question 3 part b.ii · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | arrow from top to bottom; | | 1 |
| (ii) | • A is an artery, thicker muscle wall;<br>• to pump blood from heart/to body;<br>• B is a vein with a valve present;<br>• to control direction of blood flow/to prevent backflow; | If no other marks scored award one for correct identification of both vessels | 4 |
| (iii) | capillary; | | 1 |
| (b) (i) | • decrease in deaths;<br>• in both sexes/males and females;<br>• between 2005 - 2014/over whole range; | | 3 |
| (ii) | • 60 000 000 ÷ 100 000 = 600;<br>• 600 × 640;<br>• = 384 000; | | 3 |

Total 12 marks

## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 1 of 5 · Question 3 part b.ii · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Scale a rate per 100 000 people to a whole population

Answer 2 of 5 · Question 6 part a.iii · 2 marks

2020 JAN · Paper 02 · find part a.iii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 6 (a) | (i) | • more males than females die (from *S.aureus* and MRSA)(1)<br>• number of deaths from both *S.aureus* and MRSA increase with age(1) | Allow other valid conclusions e.g more deaths from both microorganisms in males/females aged 85 and over | 2 |
| | (ii) | 416.6 ÷ 1.8(1)<br>231.4:1 (1) | Allow 231:1<br>Allow full marks for correct final answer | 2 |
| | (iii) | 41.2 + 21.9 = 63.1(1)<br>53 × 63.1 = 3344(1) | Allow 3344.3<br>Allow full marks for correct final answer | 2 |
| | (iv) | Any two from:<br>• MRSA is resistant (1)<br>• to (many) antibiotics(1)<br>• (MRSA infection) difficult to treat(1)<br>• reference to MRSA infection being (highly) contagious/out of control/easily transferred(1) | | 2 |
| (b) | (i) | mircroorganisms/bacteria/fungi | Ignore viruses<br>Ignore named antibiotics | 1 |
| | (ii) | Any six from:<br>• sterilise Petri dishes/agar/nutrient broth(1)<br>• obtain sample(1)<br>• sterile inoculating loop/sterile swab(1)<br>• pass neck of flask containing culture through a flame(1)<br>• smear culture/sample/swab onto agar(1)<br>• reference to adding antibiotics (to agar)(1)<br>• seal Petri dishes/cover with lid(1)<br>• incubate(1)<br>• measure area of clear zone around antibiotic discs(1) | Allow description of how loop/swab is sterilised | 6 |

**Total for Question 6 = 15 marks**

## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 3 of 5 · Question 4 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 4 | (a) (i) | A (bacterium);<br><br>B doesn't cause it<br>C doesn't cause it<br>D doesn't cause it | | 1 |
| | (ii) | • no of cases decreases as vaccination rate increases;<br>• lag in effect;<br>• fluctuations in number of cases;<br>• despite percentage of vaccinations remaining constant; | | 4 |
| | (b) | 2.75 per 100 000;<br><br>$\frac{2.75 \times 3\,450\,000}{100\,000}$;<br>= 95 people; | correct answer without working = 3<br>ECF = 2 | 3 |
| | (c) | • inject weakened/attenuated organism;<br>• causes antibody production;<br>• antibodies remain in blood;<br>• memory cells formed;<br>• on infection antibodies can respond quickly; | | 5 |

Total 13 marks

## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 3 of 5 · Question 4 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 4 of 5 · Question 6 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 6 | (a) (i) | • both increase;<br>• males more than females;<br>• fluctuations;<br><br>accept any two from; | | 3 marks |
| | (ii) | • layer of actively dividing cells;<br>• can become uncontrolled;<br>• when exposed to ultraviolet/sun light; | | 2 marks |
| | (b) (i) | 17 - 48;<br>= 31 per 100 000;<br>$\frac{31}{25}$ = 1.2/1.24 per 100 000; | allow 16.7 - 17 and 47.8 - 48 then ECF | 3 marks |
| | (ii) | 20 per 100 000;<br>$\frac{20 \times 7\,000\,000;}{100\,000}$<br>= 1400; | ECF | 3 marks |

Total 11 marks

## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 4 of 5 · Question 6 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Scale a rate per 100 000 people to a whole population

Answer 5 of 5 · Question 4 part c.ii · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 4 (a) | • diarrhoea by clean water/safe water/ bottled water/boiled water/ sewage treatment/chlorination/good sanitation/not mixing sewage with drinking water/washing hands/good hygiene/ cooking food thoroughly;<br>• malaria by nets/draining swamps/insecticides/ insect repellent's/wearing clothing to cover arms (and legs/mosquito coils/anti-malarial/vaccination ; | Ignore ORT | 2 |
| (b) (i) | antibiotics; | | 1 |
| (ii) | • elderly have weakened/lowered immune system/immunocompromised;<br>• indoor spaces makes transmission easier/reduced air flow/ reduced ventilation/build- up of air droplets with pathogens/ people closer together; | | 2 |
| (c) (i) | • diet;<br>• exercise;<br>• smoking:<br>• drinking alcohol:<br>• stress | Accept drugs | 2 |
| (ii) | • $\frac{5\,000\,000}{100\,000} = 50$;<br>• 50 × 445 = 22 250;<br>• 22 300;<br>***22,250 scores 2 marks***<br>***22,300 scores 3 marks*** | ***1 mark for ×445 if there are no other marks*** | 3 |
| (iii) | • use of 445 and 185;<br>• $\frac{445}{185}$ ;<br>• 2.41:1/2.405:!/2.4:1= 3 marks; | ecf from mp1<br>**if they have done the correct calculation and 2:1 is on the answer line they score** 2 marks | 3 |
| (iv) | • people live longer/elderly population;<br>• these diseases are more common in older people;<br>• less likely to die younger from infectious diseases/better medical facilities/treatment;<br>• different diet/ more meat/saturated fats/processed food;<br>• drinking more alcohol; | People live longer so there is more cancer and dementia/these diseases= 2 marks | 2 |

**Total 15 marks**

## Type MS · Scale a rate per 100 000 people to a whole population

### Answer 5 of 5 · Question 4 part c.ii · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |